

# ALI SHAHIDI

Calgary, AB · +1(403)437-4734 · [ali.shahidi.as1997@gmail.com](mailto:ali.shahidi.as1997@gmail.com)  
[linkedin.com/in/ali-shahidi97/](https://www.linkedin.com/in/ali-shahidi97/) · [github.com/Alishahidi1997](https://github.com/Alishahidi1997) · <https://alishahidi1997.github.io/>

## SUMMARY

Software Developer focused on Unity/XR systems, C#/.NET/ Python backend engineering, and applied AI. Built ASP.NET Core APIs with JWT and role-based access, delivered multiplayer VR/AR and multimodal AI systems, and used agent/automation workflows to improve delivery speed, reliability, and maintainability.

## PROFESSIONAL EXPERIENCE

**Software Developer** *September 2024 - November 2025*  
**ETHEREAL Research Group** *Calgary, AB*

- Built an accessible VR training system for non-speaking autistic users, improving interaction flow and ethical UX outcomes.
- Engineered backend services to synchronize GPT-driven dialogue, user events, and multimodal sensory feedback in real time.
- Improved video processing performance by optimizing the data pipeline and reducing redundant computation.
- Reduced API usage and cost by sending structured metadata instead of raw video streams.

**Software Developer** *September 2024 - March 2025*  
**ELIXR & Illumia** *Calgary, AB*

- Delivered VR/AR solutions across 5 projects using Unity, C#, WebXR, real-time audio, and AI-driven interaction systems.
- Built cockpit and ATC simulators with agent-style LLM orchestration connected to Python backend APIs.
- Developed RAG pipelines for context-aware querying of domain documents.
- Improved Unity runtime performance through profiling, batching, GPU instancing, and occlusion culling.

**Software Developer - Internship** *February 2024 - August 2024*  
**Telus** *Calgary, AB*

- Led development of a cross-platform multiplayer AR app for iOS and Quest, enabling real-time shared experiences for up to 50 users.
- Engineered NetCode and spatial anchor systems for accurate mapping, animation, and interactive AR workflows.
- Improved operational efficiency by supporting remote instruction using immersive AR/VR experiences, reducing the need for on-site service.

**Graduate Research & Teaching Assistant - Software Engineer** *January 2021 - April 2024*  
**University of Calgary** *Calgary, AB*

- Supported instruction in software engineering, object-oriented programming, and scalable data analysis.
- Built software prototypes and data workflows for assistive technology research with interdisciplinary collaborators.

## EDUCATION

**University of Calgary** *April 2024*  
Master of Software Engineering (GPA: 4/4.00) *Calgary, Canada*

- Received full graduate funding and an international tuition award; nominated for the Distinguished Master's Thesis Award in STEM.

**University of Tehran** *July 2020*  
B.Sc. in Computer Engineering (GPA: 17.24/20.00) *Tehran, Iran*

- Received Best Undergraduate Project and two Faculty of Engineering Fellowship Awards.

## PROJECTS

---

### **Social Networking API** | ASP.NET Core, MVC, REST APIs, MSSQL, Dapper, Entity Framework

- Built a full-stack backend using ASP.NET Core and MSSQL following MVC architecture, exposing RESTful APIs for user management with Swagger documentation.
- Implemented JWT authentication, secure password hashing, and role-based authorization for user/admin workflows.

### **ETHEREAL Research Group - Coffee Shop Simulation** | Unity, C#, .NET, Python, AI, API, VR, SQL

- Developed a high-fidelity VR training environment for neurodivergent users, integrating multimodal sensory feedback and ethical UX design.
- Engineered a Python backend for real-time state management and asynchronous communication between Unity and external LLM services.
- Integrated GPT dialogue and voice synthesis for adaptive NPC interactions and rubric-style trainee feedback.

### **ETHEREAL Research Group - Multi Modal Behaviour Analysis** | Python, AI, Unity, VR/AR

- Built a multimodal behavior analysis system (Meta Quest 3 + Python) combining vision, audio, and speech for real-time event detection.
- Designed a window-based pipeline using MediaPipe pose (keypoints + visibility), motion heuristics, and majority voting for robust behavior detection.
- Implemented audio pipeline with WhisperX; including diarization, aligning speech, speaker roles, and loudness (dB) with video frames.
- Developed a privacy-first architecture: on-device feature extraction with structured outputs sent to GPT for behavioral interpretation and natural-language summaries.

### **ELIXR & Illumia - Hybrid RAG Retrieval** | Python, FAISS, BM25, RAG, JSONL, FastAPI

- Built a hybrid RAG system over a curated aviation corpus (regulations, flight procedures, communications, weather), combining FAISS (dense) and BM25 (lexical) with Reciprocal Rank Fusion (RRF) to improve retrieval for both semantic queries and exact-match terms.
- Designed an end-to-end retrieval pipeline (retrieve → RRF fusion → cross-encoder rerank → LLM generation) to produce grounded answers from domain documents.
- Structured unstructured PDFs into chunked JSONL and maintained FAISS + BM25 indexes; implemented LRU embedding caching to reduce redundant API calls and improve latency.
- Developed a lightweight evaluation pipeline using Recall@K, Precision@K, and MRR to measure and iterate on retrieval quality.
- Exposed the system via a FastAPI endpoint for real-time querying through a REST API.

### **ETHEREAL Research Group - Internal Backend API** | FastAPI, JWT, SQLite, OpenAI API

- Developed a FastAPI backend to support internal research workflows, implementing JWT-based authentication and user-scoped APIs for task tracking, summaries, and analytics.
- Built scheduled background jobs (APScheduler) to aggregate task activity and generate per-user daily summaries for progress monitoring.
- Integrated AI-powered endpoints for task parsing and insight generation with fallback logic to ensure consistent system behavior.

### **Community Sim** | Python, APIs, JSON, Automation

- Developed and maintained Community Sim in Python, integrating API workflows and structured JSON data exchange.
- Built agent/skill-driven automation workflows for setup and validation, improving release consistency and reducing manual overhead.

## SKILLS

---

**Languages:** C#, Python, C, C++, JavaScript, SQL

**AI & Machine Learning:** RAG, FAISS, BM25, embeddings, Llama 3, OpenAI API, agent/skill workflows, multimodal inference, NLP

**XR & Systems:** Unity, Meta Quest SDK, WebXR, NetCode (Multiplayer), Spatial Anchors, AR Foundation

**Backend & Cloud:** ASP.NET Core, MVC, RESTful APIs, FastAPI, Docker, Micro-ORMs (Dapper)

**Databases:** MS SQL, MySQL, PostgreSQL

**Tools & Workflow:** Git, Linux, LaTeX, Cursor, Claude, Automation, Scrum/Agile, Jira, Confluence